Numbers, Percents & Ratios

Section 1: Number Properties, Absolute Value & Order of Operations

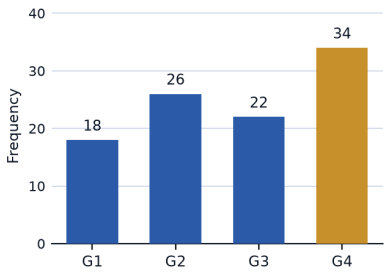
1 What is the value of the numerical expression $|-14|-3|5-9|+(-2)^3$?

a. -6

b. -10

c. 2

d. 14



2 If p is a positive prime number and q is a negative integer, which of the following values must always be negative?

a. $p-q$

b. $q-p$

c. p^2+q^2

d. $\frac{p}{q^2}$

3 What is the greatest common factor of $72, 108,$ and 180?

a. 18

b. 24

c. 36

d. 72

Section 2: Proportions, Rates & Scale Conversions

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4 On a blueprint, a scale of $\frac{1}{4}$ inch represents 3 feet. If a conference room has a length of 36 feet, what is its length on the blueprint?

a. 2.5 inches

b. 3 inches

c. 4.5 inches

d. 6 inches

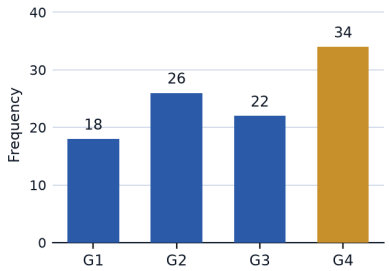
5 A car travels 180 miles on 6 gallons of fuel. At the same rate of fuel consumption, how many gallons of fuel are required to travel 450 miles?

a. 12

b. 15

c. 16.5

d. 18



6 The ratio of copper to zinc in a brass alloy is $5:3$ by weight. If a piece of brass weighs 56 kilograms, how many kilograms of copper does it contain?

a. 21

b. 30

c. 35

d. 40

Section 3: Percentages, Discounts & Successive Changes

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7 A store purchases a bicycle for 150 and marks up the price by 40%. During a weekend clearance sale, the bicycle is discounted by 20%. What is the final sale price?

a. 168

b. 174

c. 180

d. 192

8 If 35% of a number N is equal to 84, what is 60% of N?

a. 120

b. 144

c. 168

d. 240

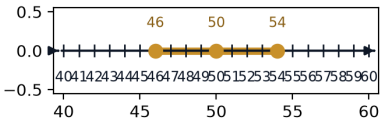
9 The price of an airline ticket increased from 250 to 320. What was the percent increase in the ticket price?

a. 22%

b. 25%

c. 28%

d. 30%



Section 4: Scientific Notation, Exponents & Radicals

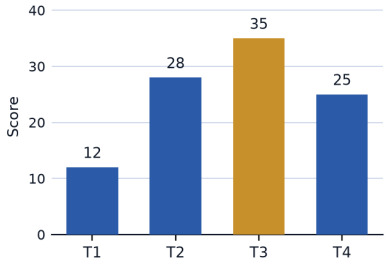
10 What is the value of $(3.2 \times 10^{7}) \times (2.5 \times 10^{-3})$ expressed in standard scientific notation?

a. 8.0×10^{4}

b. 8.0×10^{5}

c. 8.0×10^{3}

d. 0.8×10^{5}



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11 Which of the following is equivalent to $\frac{x^5 \cdot x^{-2}}{(x^3)^2}$ for all positive values of x?

a. x^{-3}

b. x^{-1}

c. x^0

d. x^2

12 Simplify the radical expression $\sqrt{75} - 2\sqrt{27} + \sqrt{48}$.

a. $2\sqrt{3}$

b. $3\sqrt{3}$

c. $4\sqrt{3}$

d. $5\sqrt{3}$

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